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Sentiment Analysis on crypto related comments using DistilBERT

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Chapter 1

Introduction

In this project we will conduct a sentiment analysis (sometimes referred also as *opinion mining*) of comments made by users of the crypto community using the DistilBERT model. The aim of the project is to determine the effectiveness of fine tuning on a specific domain such as the crypto community, which is rich of peculiar terms and idioms commonly used by its users.

The fine tuning process was applied to a Transformer which was previously pre-trained on the Stanford Sentiment Treebank v2 (SST2) which is a standard dataset for sentiment analysis on movie reviews. The Transformer model used was the DistilBERT SST2 which was provided by the Python library `transformers` from HuggingFace. This library allows to load various pre-trained models, import labeled datasets and also to use a useful API which ease the work on NLP tasks. The labeled dataset used for the training and testing of DistilBERT was provided by SocialGrep which was responsible for the collection, aggregation and labeling of the comments taken from various subreddits (i.e. forums on Reddit) about crypto for the period of August 2021. A second unlabeled dataset with comments coming exclusively from the "r/cryptocurrency" subreddit was used to conduct experiments on the model. This dataset was built using the PushShift API for Reddit which is used to collect data of posts and comments from Reddit and pmaw which is a wrapper for the PushShift API to make multiple requests.

Finally we experimented with this unlabeled dataset to try to understand the correlation between the price of Bitcoin (the most popular cryptocurrency) and the sentiment of the community, which we believed to be a strong one. The results demonstrated that in various crashes of the market during the month of November 2021 the sentiment score dropped significantly with respect to the previous days. Thus showing that a strong price action of Bitcoin and more general of cryptocurrencies implies a strong response in sentiment of the community.

Chapter 2

Cryptocurrency community

A cryptocurrency is defined as a digital currency designed to work as a medium of exchange through a computer network that is not reliant on any central authority, such as a government or bank, to uphold or maintain it.

The first and most popular crypto is by far Bitcoin, which was invented in 2008 by an unknown person or group of people using the name Satoshi Nakamoto[1]. The vision of Satoshi was to have *"A purely peer-to-peer version of electronic cash [which] would allow online payments to be sent directly from one party to another without going through a financial institution"*. In this way no centralized party had to be involved in the exchange of goods or for pure payments, this was achieved by using cryptography and a peer-to-peer network which is able to validate the transactions through a Proof of Work system. A public ledger records all bitcoin transactions and copies are held on servers around the world. Anyone with a spare computer can set up one of these servers, known as a node. Consensus on who owns which coins is reached cryptographically across these nodes rather than relying on a central source of trust like a bank.

Figure 2.1: Bitcoin and Ethereum

The introduction of Bitcoin and more in general of crypto, have been vital to a lot of people who couldn't rely on the classical banking and financial system. Further, Bitcoin proved to be, not just a medium of exchange, but also a deflationary store of value which could prevent the effects of the constant rising inflation of traditional fiat currency (e.g. USD, EUR).

The role of crypto reflected also in our society and culture giving importance to the decentralized aspects of our data, one such example is the introduction of the so called "Web 3.0", an idea for a new iteration of the World Wide Web based on blockchain technology, which incorporates concepts such as decentralization and token-based economics. One of the greatest revolution in the crypto sphere was the introduction of smart contracts, a transaction protocol which is intended to automatically execute, control or document legally relevant events and actions according to the terms of a contract or an agreement. Ethereum has been one of the first cryptocurrency to integrate smart contracts in their blockchain by leveraging the EVM (Etherem Virtual Machine), a Turing Complete state machine which allows to run code written for example in Solidity, which is the most popular programming language using the EVM.

The Bitcoin code is open-source and it is powered by the community which proposes and discusses new features and also maintains the source code, the whole ecosystem thrives on the actual efforts of developers and enthusiasts, this is one of the reason why decentralization has been a prominent feature. The community does not just affect the development of a cryptocurrency but it also influences its price, one of the main factors is the popularity behind the coin or in some cases how much the investors participate in the protocol. Communities are often gathered in online social spaces such as Twitter, Discord or Reddit. In our experiments we will use datasets which comprehend comments coming from subreddits such as: "r/CryptoCurrency", "r/SatoshiStreetbets", "r/CryptoMoonshots" and many others. In these social spaces people often write and discuss about crypto news or the general state of the crypto market, thus comments collected from this subreddits seem ideal to analyze the sentiment of the crypto market.

Chapter 3

Sentiment Analysis and DistilBERT

Sentiment analysis is the task of classifying the polarity of a given text at the document, sentence, or feature/aspect level, whether the expressed opinion in a document, a sentence or an entity feature/aspect is positive, negative, or neutral. This kind of classification task is commonly used for analyzing reviews, survey responses, online and social media to have an insight on a particular product or idea.

In this project the text analyzed will be comments coming from a crypto forum where users participate in discussions about crypto news, ideas, analysis and more in general the state of the crypto ecosystem. The comments are taken from an already labeled dataset from SocialGrep which encompasses a month's worth of posts and comments from selected cryptocurrency subreddits, indexed for the whole month of August 2021, in order to preserve users' anonymity and to prevent targeted harassment, the data does not include usernames. This first dataset used for training and testing had a total of roughly 3.7 million rows which have been reduced to almost 2 million rows, the preprocessing phase deleted comments that had a "[deleted]" or "[removed]" body, comments that were made by bots and also comments which were longer than 512 characters. This dataset was composed by the *body* of the comment, which is the text collected and by the *sentiment* which is a value between 0 (negative) and 1 (positive).

The actual number of rows used for training was one hundred thousand, whereas for the evaluation of the fine tuning and the testing of the final model we used twenty-five thousand each. This is due to the great amount of time that the fine-tuning process takes to train a model with so many parameters, so for this reason we chose to just use a small subset of the data available.

The DistilBERT model is a Transformer model which uses a mechanism called *Attention* first introduced by Google Brain in 2017[2]. DistilBERT was proposed in the blog post *"Smaller, faster, cheaper, lighter: Introducing DistilBERT, a distilled version of BERT"* and the paper *"DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter"*[3]. This model is a small, fast, cheap and light Transformer model trained by distilling

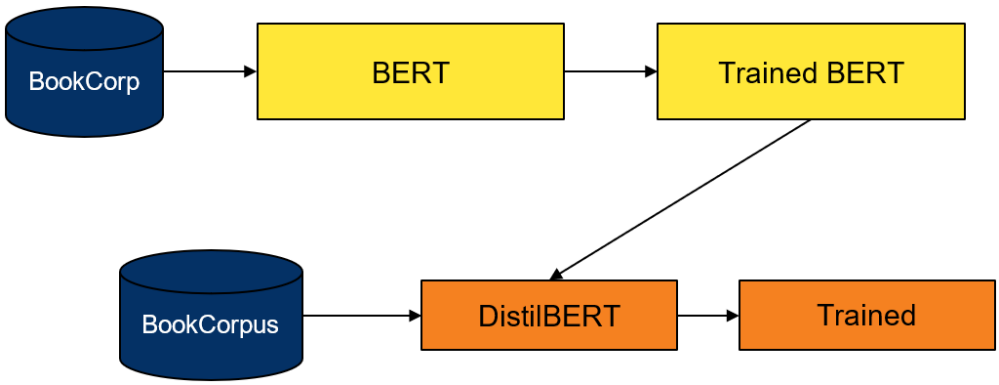

Figure 3.1: The DistilBERT model

BERT base. It has 40% less parameters than the BERT model, runs 60% faster while preserving over 95% of BERT's performances as measured on the GLUE language understanding benchmark. The distillation process is a compression technique in which a compact model (the student) is trained to reproduce the behaviour of a larger model (the teacher) or an ensemble of models, in DistilBERT the teacher is the BERT model which is bigger in size, with 110 million parameters whereas the distilled version has only 66 million parameters. Training a massive language model (such as BERT) learns millions of latent features which are then picked by the finetuning process improving them over the domain provided (in our case the crypto related discussions). This requires an oversized model, because only a subset of the features are useful for any given task , distillation allows the model to only focus on those features.

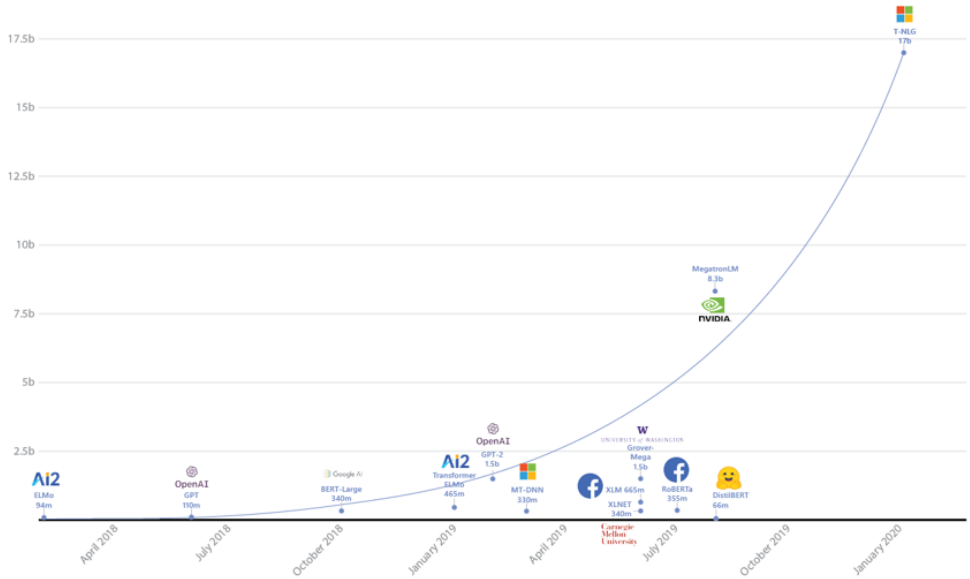

Figure 3.2: Number of parameters of various Transformers models

Chapter 4

Experiments

First of all we preprocessed the dataset by removing irrelevant columns and comments (e.g. comments made by bots) from the dataset, then the dataset needed to be converted to the `Dataset` format accepted by the HuggingFace model. Next, the dataset had to be tokenized using the tokenizer associated with the DistilBERT SST2 model, in this way the data was ready for the experimental phase.

The first experiment was to try to predict the polarity of the comments in the test set using the pre-trained model to establish the accuracy without finetuning, in this way we provided a baseline to analyze the improvement with the finetuning process. HuggingFace provides the `pipelines` library which are an easy way to use models for inference, these pipelines are objects that abstract most of the complex code from the library, offering a simple API dedicated to several tasks, e.g. Named Entity Recognition, Masked Language Modeling, Sentiment Analysis, Feature Extraction and Question Answering. The inference on the test set using the standard pre-trained model DistilBERT SST2 took more than one hour [1] using the pipeline implementation. The result is that the model provided an accuracy of only 63% accuracy against the reported accuracy of 91.3% on the development set on the SST2 dataset. This low accuracy achieved shows how the pre-trained model performs poorly on text which presents a specific vocabulary that the model has unlikely seen. In fact the hypothesis is that the model has a hard time trying to classify correctly the comments due to the fact that common terms in the crypto community such as "bullish" which is often addressed as optimistic and confident feeling about the market are not well known to the model, so fine tuning will increase the chance of a better understanding of the polarity of the comments.

Next we started the finetuning with the training set (made of one hundred thousand comments), the training process was done using the `Trainer` class provided by HuggingFace which ease the training process of the Transformer model. In this case the pre-trained model needed to be loaded and subsequently the training set

[1] All the experiments have been conducted using a Nvidia RTX 3070 GPU

Model	Accuracy	Precision (avg)	Recall (avg)	F-measure (avg)
Pre-trained DistilBERT	63%	67%	71%	62%
Fine-tuned DistilBERT	96%	95%	95%	95%

Table 4.1: Summary of the results achieved

was provided to start the training which took roughly six-eight hours to complete three epochs. Finally the predictions made based on the test set (made of twenty-five thousand comments) returned a result of 96% accuracy. We can establish that 33 percentage points improvement was a significant one, which now made the model able to predict the polarity of the comments in a correct way.

The next experiment was to compare the sentiment of the community with the price of Bitcoin, the hypothesis is that a great fall in price for Bitcoin implies also a fall in the sentiment of the community. In this case we used a second unlabeled dataset with 500.000 comments taken only from the "r/CryptoCurrency" subreddit which is the most used forum of all the mentioned before. The timeframe selected for the comments in the dataset is the month of November 2021, we believe this timeframe to be significant because of multiple sudden crashes of the crypto market which happened throughout the month. The dataset has been preprocessed exactly as before by deleting columns and rows which were irrelevant, in this case there was no "sentiment" column since the dataset has just been collected with no labels. The only difference is that the data has been partitioned according to the days, in such way we obtained multiple groups (dataframes of data grouped by day). For each day we predicted the sentiment of each comment classifying it as positive (1) or negative (0) and then we computed a sentiment score which reflected the overall sentiment of the day as $score_i = \frac{count(pos_i)}{count(neg_i)}$ where $i \in [1, 30]$.

Next we retrieved the prices of Bitcoin for the month of November using the CoinGecko API, we obtained a price value for each single day. Finally we plotted the prices of Bitcoin and the sentiment scores of November which are showed in figure 4.1. We can observe that even though there is a somewhat of correlation between the two sequences there is not a real correspondence. This aspect is due to the fact that the price of Bitcoin is very volatile during the single day, so a single value for the day does not really capture the continuous falls and rises of the prices. For this reason we also show a more representative plot which is the candlestick graph of the Bitcoin price. Since it is believed that markets are strongly affected by the emotion of traders, in a phenomenon known as "emotional trading"[4], these candlesticks show that emotion by visually representing the size of price moves with different colors. The comparison of the candlestick graph with the sentiment trends (see figure 4.2) shows clearly how the great sudden crashes on days 10, 15 and 25 affected the sentiment of the community. This further proves the correctness of the approach and how the model is able to capture the sentiment of the community.

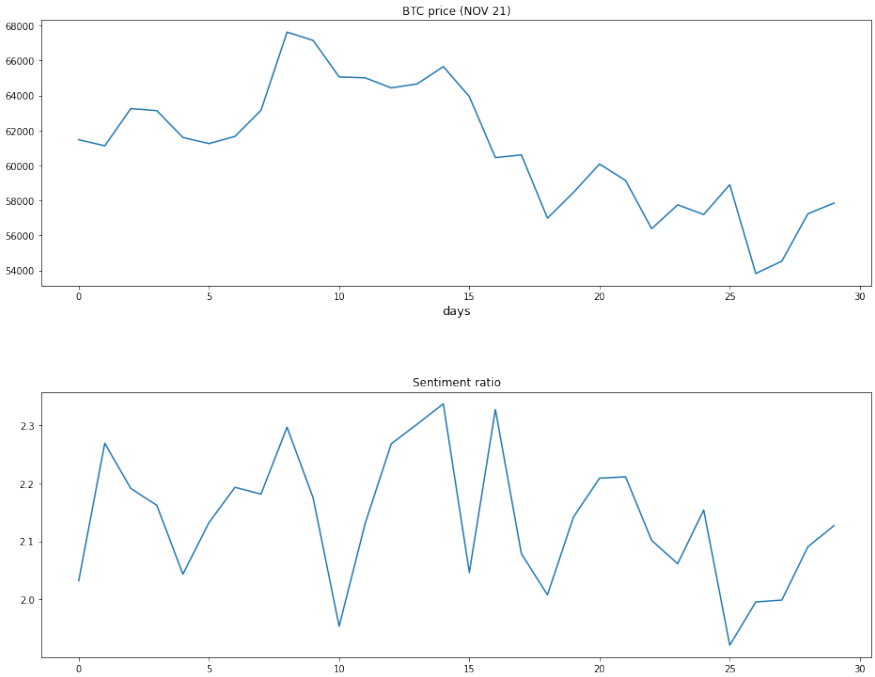

Figure 4.1: Correlation between Bitcoin price and sentiment score

Figure 4.2: Correlation between Bitcoin price (candlesticks) and sentiment score

Chapter 5

Conclusions

We used a labeled dataset of comments made by the crypto community to perform fine tuning on the DistilBERT SST2 pre-trained model provided by HuggingFace. In this way we improved the accuracy from 63% (pre-trained model) to 96% (fine-tuned model) on the test set. We tried to explain the poor results on the pre-trained model as a lack of context for the domain specific vocabulary. We also showed that the fine-tuning process helped understand the context of comments made about crypto and the way of some terms are commonly used. Finally we showed a practical example of how to analyze the movements of the crypto market in comparison to the sentiment showing that the approach is well-founded. A further work that could be done is an analysis on a much longer span and possibly on multiple sources of data, this could possibly show if a so called "wisdom of the crowd" could predict rises or falls of the market. More in general the model could help in finding some kind of insight that could be extrapolated from the sentiment of the community.

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